MD. EASIR RAHAT

Mechanical Engineering Graduate | Machine Learning for Thermal & Mechanical Systems
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PORTFOLIO

Research Objective

Mechanical Engineering graduate (RUET, 2026) seeking a PhD position in **machine learning for thermal and mechanical systems**: data-driven and physics-informed models for indoor thermal comfort, building HVAC, heat transfer, and machine condition monitoring. My undergraduate thesis deploys an ensemble neural network for real-time thermal comfort prediction on embedded hardware, and my independent studies cover physics-informed neural networks and vibration-based fault diagnosis. I aim to develop ML methods that respect physical constraints and to validate them on real systems.

Education

Rajshahi University of Engineering & Technology (RUET) — **Rajshahi, Bangladesh**
B.Sc. Engineering in Mechanical Engineering | CGPA: 3.33/4.00 — Mar 2022 – Aug 2026

- **Undergraduate Thesis:** Development of an ANN-Based Indoor Thermal Comfort Prediction and Monitoring System. Advisor: Dr. Mhia Md. Zaglul Shahadat, Professor, Dept. of ME, RUET.
- **Relevant Courses:** Thermodynamics, Fluid Mechanics, Heat Transfer, HVAC & Refrigeration, Control Systems, Machine Design, Manufacturing Processes, Production Planning & Control.

New Govt. Degree College, Rajshahi (HSC, 2018 – 2020), GPA 5.00/5.00 | **Joypurhat R.B. Govt. High School** (SSC, 2016 – 2018), GPA 5.00/5.00

Research Interests

- **Primary:** Machine learning for thermal comfort, HVAC, and building energy systems; physics-informed machine learning for heat transfer; condition monitoring and predictive maintenance.
- **Methods:** Deep learning (including parameter-efficient continual learning), Bayesian optimization, embedded/edge deployment of ML models.

Publications (Submitted)

[S-1] **Md. Easir Rahat**, Md. Ferdaous Al-Farabe. "Development of an ANN-Based Indoor Thermal Comfort Prediction and Monitoring System." Submitted to the 29th International Conference on Computer and Information Technology (ICCIT 2026), Cox's Bazar. An ensemble ANN predicting indoor thermal comfort in real time, benchmarked against Random Forest, XGBoost, and Linear Regression, validated against the CBE Thermal Comfort Tool, and deployed on ESP32 hardware with a live monitoring dashboard. [Link]

[S-2] **Md. Easir Rahat**, Md. Tajbir Hasan Shuvo. "Revisiting Design Choices in LoRA-Based Continual Learning for Vision Transformers in Semantic Segmentation: An Empirical Study." Submitted to ICCIT 2026, Cox's Bazar. Reimplements a LoRA-based continual learning pipeline (CLoRA) on a frozen ViT-B/16 and varies backbone pretraining, decoder depth, distillation weight, and LoRA rank across four PASCAL VOC splits to isolate which design choices drive performance and forgetting resistance. [Link]

Independent Research Projects

Bearing Fault Diagnosis: Cross-Fault-Size Generalization (CWRU) [Code]

- Tested whether classifiers that look accurate on CWRU hold up when fault size changes: leave-one-load-out accuracy was 100%, but macro-F1 fell to roughly 0.25–0.57 when training on one fault size and testing on the other.
- Added envelope-spectrum features at bearing fault frequencies (BPFO, BPFI, BSF): Random Forest macro-F1 rose from 0.26 to 0.68 (0.014 in $\rightarrow$ 0.007 in) but not in the reverse direction (10 seeds). Documented limits (one recording per condition, two fault sizes) and built a Streamlit workflow demo.

Physics-Informed Neural Network for 1D Heat Conduction [Code]

- Trained a PINN on the transient heat equation from the PDE and boundary/initial conditions alone (0.31% relative L_2 error); benchmarked against explicit FTCS and Crank–Nicolson solvers, which matched its accuracy in about 0.3 ms versus 42–65 s of training, so the PINN is not competitive as a forward solver.
- Recovered an unknown thermal diffusivity from 30 noisy measurements (0.16% error on clean data, 2.9% at noise std 0.05). Single seed; a classical inverse baseline is the planned next comparison.

Bayesian-Optimization Car Setup Tuner [Code] [Demo]

- Built a Gaussian-process optimizer (Expected Improvement) from scratch in JavaScript over a quasi-steady-state lap-time model; across 20 trials of 30 runs it reached 77.7 s versus 79.3 s for random search.

Animal Image Classifier (MobileNetV2 Transfer Learning) [Code]

- 97.4% accuracy and 0.971 macro-F1 on 2,619 held-out images (about 26,000 images, 10 classes). A paired McNemar test ($p = 0.12$) showed the improved training recipe's gain was within noise; also found and fixed an evaluation misalignment bug and identified dataset labeling errors.

Skin-Condition Classification on SCIN: A Protocol-Logged Pilot with Controls [Code]

- Designed a protocol-first study on Google's SCIN dermatology dataset (research prototype, not for diagnosis) with group-aware splits (StratifiedGroupKFold over duplicate-linked cases), pooled out-of-fold metrics with group-bootstrap 95% CIs, and metadata-only and class-weight controls. The head/neck pilot (114 cases, 4 classes) was a null result, so scope moved to an exploratory all-body-parts study (1,172 cases, 8 classes) after a manual face review of 267 images.
- Fine-tuned MobileNetV3-Small reached 0.301 balanced accuracy (chance 0.125) but was not distinguishable from a weight-matched metadata baseline, and fine-tuning was not shown to beat frozen features. Unweighted models were overconfident; confidence-based abstention helped only partly. Darker skin tones (FST5–6) were 6.6% of cases, so no fairness claim is made.

Industrial Training

Industrial Trainee, Engineering Department **10 – 23 Mar 2025**
Square Pharmaceuticals PLC, Pabna Unit, Bangladesh ME 4120; Supervisor: Md. Khairul Alam, DGM

- Two-week rotation across plant utilities: the HVAC system for cleanrooms and warehouses (air handling units, chillers, dehumidifiers, Building Management System), a 17-compressor compressed air network, and the 11/0.4 kV substation fed by a 9 MW captive gas-turbine plant.
- Studied purified water and Water-for-Injection systems, nitrogen/oxygen generation (molecular-sieve adsorption), and QC testing against BP/USP/EP standards; the final report raised observations on manual packaging and warehouse capacity.

Engineering Projects

Go-Kart Quality Control Lead, Team Annexe RUET Aug 2025 – Aug 2026
Formula Kart Design Challenge, Coimbatore, India

- Checked incoming materials and dimensions against design tolerances, tracked fit issues during welding and assembly, and sourced components on a tight budget without compromising safety-critical parts (steering, brakes, roll structure); ANSYS was used for structural checks on the chassis before fabrication.

Simulation Studies (ANSYS Static Structural, SolidWorks Flow Simulation)

- Brake/lever assembly under a 300 lbf load (von-Mises stress, displacement); spur gear pair contact stress under 274 lbf·in torque; internal flow in a multi-bend piping loop (temperature, vorticity, boundary layer development).

Mechanical and Embedded Builds

- Three-axis robotic arm (Arduino Uno, three servos, independent push-button control per joint, circuit tested on a breadboard); object-pulling crane prototype (linkages and load handling); AutoCAD 2D/3D modeling and drawing of mechanical parts.

Skills

- **ML & Programming:** Python, PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LoRA/PEFT, Streamlit, MATLAB, JavaScript
- **Embedded:** ESP32, Arduino (sensor deployment and live dashboards)
- **Thermal & Mechanical:** HVAC & Refrigeration, Heat Transfer, CFD Fundamentals, Machine Design; ANSYS, SolidWorks, AutoCAD
- **Quality Tools:** 5S, Six Sigma, Root Cause Analysis, 5M, Fishbone Diagram

Leadership

Vice President (Finance), ASHRAE RUET Student Branch Jul 2025 – Aug 2026

- Coordinated two technical events with the ASHRAE Bangladesh Chapter, including a two-day HVAC Design seminar with 150+ participants; managed branch budgeting and resources.

Vice President (Program), RUET Mechanical Club 2025 – 2026

- Organized the "Thesis Roadmap & Case Study Competition," a department-wide event on research planning and applied case analysis.

Class Representative, Dept. of Mechanical Engineering, RUET Oct 2023 – Aug 2026

– Liaison between faculty and 180+ students across three sections for over two years; organized the farewell, reception, and industrial tour.

General Secretary, Joypurhat Zilla Association of RUET Jun 2025 – Jul 2026

– Organized events for 50+ members, ran admission booths for new students, and organized a Science Olympiad for school students in Joypurhat.

Certifications

HVAC Design Session, Certificate of Attendance (10 PDH), ASHRAE RUET Student Branch, Jan 2026 [Certificate] | Industrial Training Certificate, Square Pharmaceuticals PLC, Mar 2025 [Certificate]

References

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